Analog Quantities → continuous in nature.
↳ can take in any values in a Specific Range.
e.g Intensity of light, temp.

Digital Quantities → uses discrete values / Specific values.
e.g time on clock

Sampling of analog graph (taking Specific values at certain interval) generates a digital graph

Digital System:. → process information in discrete values and not in continuous form
↓ (Less prone to noise)
not analog
e.g calculator, 0, 1 bits.



to process analog data IN A DIGITAL SYSTEM we use ↓ these two.

1) Sampling at regular time intervals
2) Quantization → dividing Y-axis into Small Intervals

Binary Value Quantization → less prone to noise
↳ But requires more wires to carry

Conversions: possible digits

Binary (0 - 1)
Octal (0 - 7)
Decimal (0 - 9)
Hexadecimal (0 - 9, A - F)

For valid representation digits must be less than the base

0 1 1 1 1 0 1 1 1 0 0 0 0 1 0 1 0 1 1 → LSB
↓
MSB

A	10
B	11
C	12
D	13
E	14
F	15

Binary → decimal

$(11010)_2 \rightarrow (\ )_{10}$
$= 26$

16	8	4	2	1
1	1	0	1	0

$16 + 8 + 2 = 26$

Decimal → Binary

$(115)_{10} \rightarrow (\ )_2$

2	115
2	57 - 1
2	28 - 1
2	14 - 0
2	7 - 0
2	3 - 1
	1 - 1

$$2\overline{)115}\ 57$$
$$-114$$
$$1$$

$$2\overline{)57}\ 28$$
$$-56$$
$$1$$

$(1110011)_2$

Hexadecimal ⟶ Binary

$(4F91)_{16}$ ⟶ ( )2

Making 4 groups

4	F	9	1
8 4 2 1	8 4 2 1	8 4 2 1	8 4 2 1
0 1 0 0	1 1 1 1	1 0 0 1	0 0 0 1

⟶ $(0100\ 1111\ 1001\ 0001)_2$

Binary ⟶ hexadecimal

$(0100\ 1111\ 1001\ 0001)_2$ ⟶ ( )16

8 4 2 1
0 1 0 0
= 4

8 4 2 1
1 1 1 1
= 15 = F

8 4 2 1
1 0 0 1
= 9

8 4 2 1
0 0 0 1
= 1

$(4F91)_{16}$

Octal → Binary

$(327)_8 \rightarrow (\quad)_2$ making groups of three

4 2 1	4 2 1	4 2 1
0 1 1	0 1 0	1 1 1

$\rightarrow (011\,010\,111)_2$

Binary → octal (Same But Reverse)

Decimal →

- ☆ Binary
- ☆ hexa
- ☆ octal

USE



Any other Bases → Decimal

$(B46)_{16} \rightarrow (\quad)_{10}$

$(10\times16^2 + 4\times16^1 + 6\times16^0)$

→ multiply every element with the given Base inorder to get final Result in decimal.

Including fractions

$(3BA4)_{16} \longrightarrow (\quad)_{10}$

$(3\times16^3 + 11\times16^2 + 10\times16^1 + 4\times16^0)$

$= (15268)_{10}$

$(390)_{10} \longrightarrow (\quad)_{16}$

16	390
16	24 - 6
	1 - 8

$(186)_{16}$

$$16\overline{)390}\ 24 \quad (-384) \quad 6$$

$$16\overline{)24}\ 1 \quad (-16) \quad 8$$

convert $(123)_4 \rightarrow (\quad)_{10}$

$(1\times4^2 + 2\times4^1 + 3\times4^0) = (3+8+16)$

$= (27)_{10}$

convert $(1254)_{10} \rightarrow (\ )_3$

3	1254
3	418 - 0
3	139 - 1
3	46 - 1
3	15 - 1
3	5 - 0
	1 - 2

$3\overline{)1254}$

$$\begin{array}{r} 139 \\ 3\overline{)418} \\ -417 \\ \hline 1 \end{array}$$

$(1201110)_3$

FRACTIONS :

In any number System Base a.k.a Radix (r).

$(\text{fractions})_{\text{Anybase}} \rightarrow (\ )_{10}$

$(11010.11)_2 \rightarrow (\ )_{10}$

$(1\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 0\times2^0) + (1\times2^{-1} + 1\times2^{-2})$

$= (2 + 8 + 16) + (\frac{3}{4})$

$= (26.75)_{10}$

Converting fractions from decimal → other bases

$(24.25)_{10} \rightarrow (\quad)_2$

Solve integer part seperately

2	24
2	12 - 0
2	6 - 0
2	3 - 0
2	1 - 1

$(11000)_2$

Fractional part

$0.25 \times 2 = 0.5$ 0

$0.5 \times 2 = 1.0$ 1 take away whole no.

$0.0 \times 2 = 0$

Keep multiplying until result is zero

$0.25 = 01$

$(11000.01)_2$

Binary addition

$0 + 0 = 0$

$0 + 1 = 1$

$1 + 1 = 0$ carry 1

$1 + 1 + 1 = 1$ carry 1

Binary Subtraction:

$$\begin{array}{r} 0 \\ -\,0 \\ \hline 0 \end{array} \qquad \begin{array}{r} 1 \\ -\,0 \\ \hline 1 \end{array} \qquad \begin{array}{r} 1 \\ -\,1 \\ \hline 0 \end{array}$$

Borrowed

$$\begin{array}{r} 1\,0 \\ 1 \\ \hline 1 \end{array} \quad 2 - 1$$

$$\begin{array}{r} {}^{0}\not{1}\,{}^{1}\not{1}\,{}^{1}\not{0} \\ -\,0\,1\,0\,1 \\ \hline 1\,0\,0\,1 \end{array} \Rightarrow (1001)_2$$

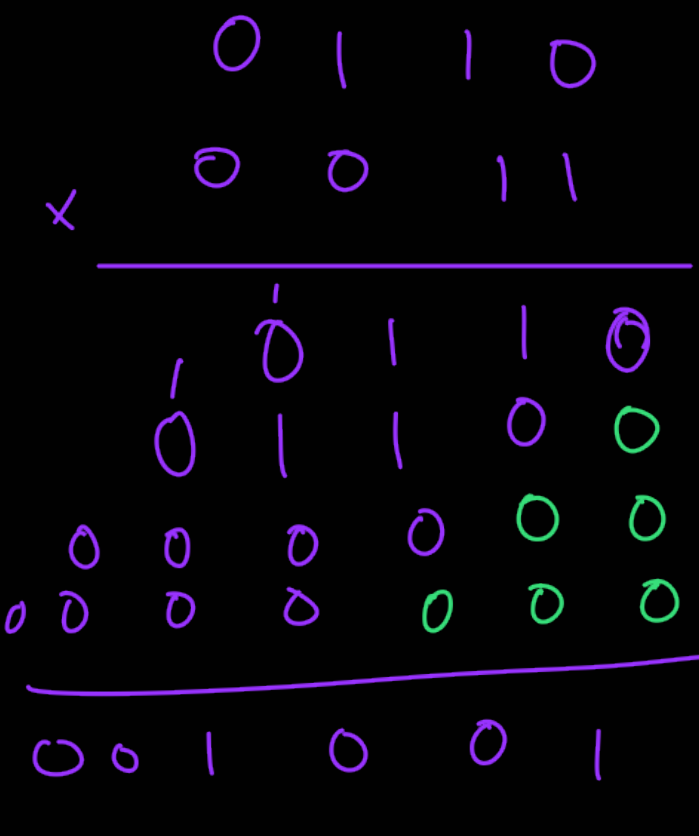
$$\begin{array}{r} 0\,{}^{0}\not{1}\,{}^{1}\not{0}\,{}^{1}\not{0}\,0 \\ -\,0\,1\,1\,1\,1 \\ \hline (0\,1\,0\,0\,1)_2 \end{array}$$

$$\begin{array}{r} 0\,{}^{0}\not{1}\,{}^{1}\not{0}\,{}^{1}\not{0} \\ -\,0\,1\,0\,1 \\ \hline 0\,1\,1\,1 \end{array} \Rightarrow (0111)_2$$

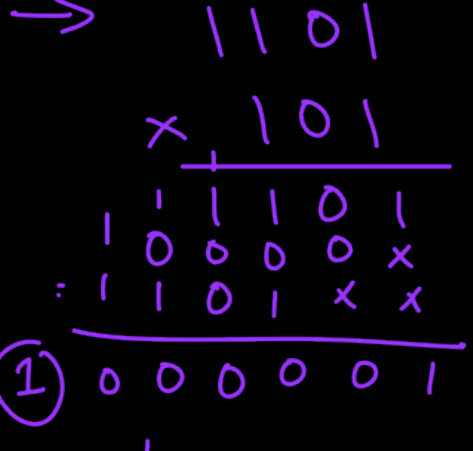
$$\begin{array}{r} {}^{0}\not{1}\,{}^{1}\not{0}\,{}^{1}\not{0}\,0 \\ -\,0\,0\,0\,0\,1 \\ \hline (1\,0\,1\,1\,1)_2 \end{array}$$

Binary multiplication

Just normal multiplication.



$\rightarrow (10010)_2$



$\rightarrow (1000001)_2$

For decimal point. Just count the no. of dp. And add it in your final Answer

multiplication by left Shifting

10110 x 1101 = ?

```
      1 0 1 1 0
    x   1 1 0 1
  -------------
      1 0 1 1 0
    1 0 0 0 0 x
  1 1 0 1 1 0 x x
  1 0 1 1 0 x x x
  -------------
  1 0 0 0 1 1 1 1 0
```

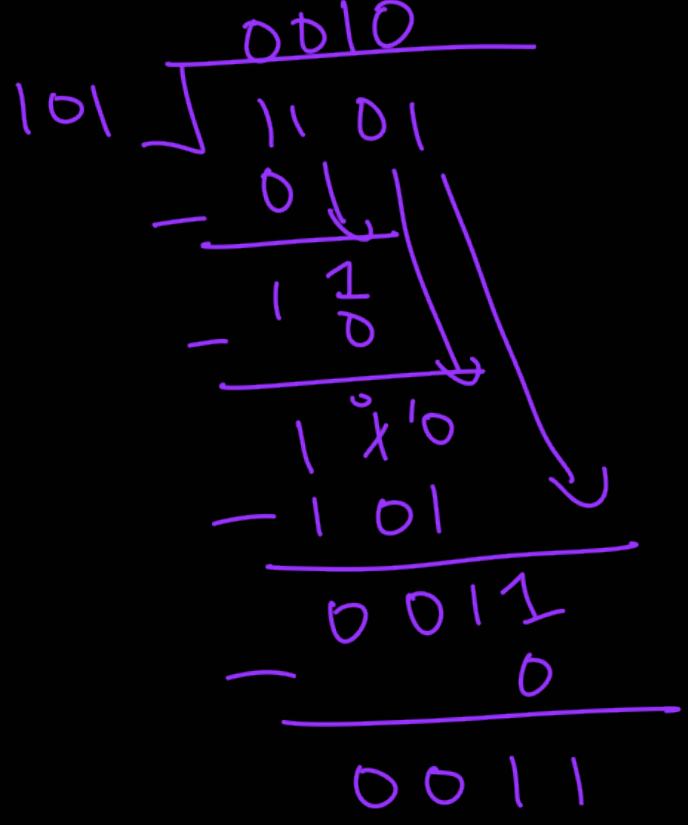


Addition of digit brought down then add zeros

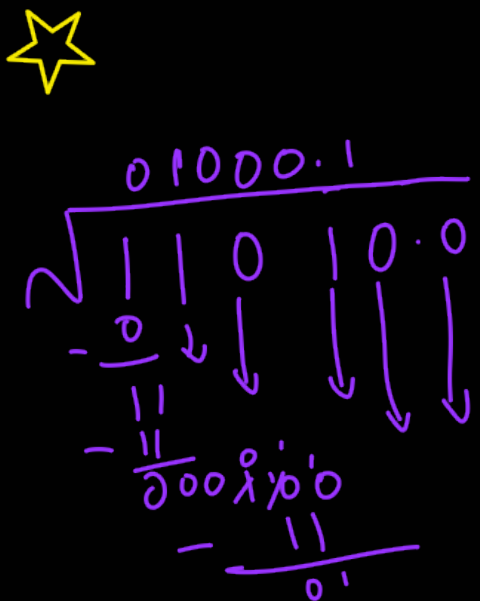
Binary division

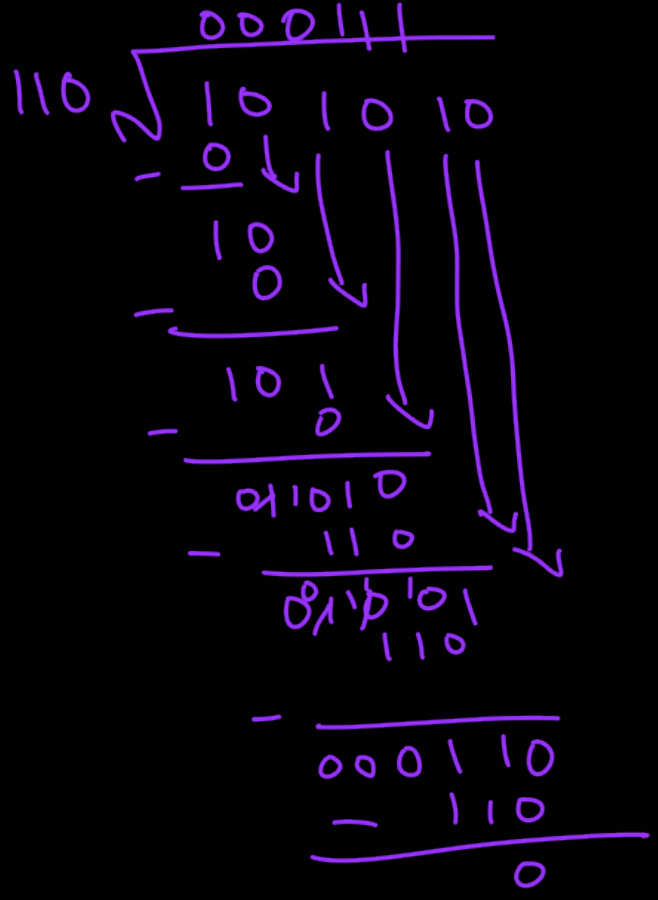
(After each Subtraction bring 1 number down)

Only two possibilities either 0 or 1



110 > 101





Complement

↓

used for Subtraction

Signed Integers MSB = 1 (negative)
= 0 (positive)

negative integers are represented using 2's complement

Two types of complement

1) Radix complement (r) a.k.a 2's

2) Diminished radix complement (r-1)
a.k.a 1's

Dimished complement of $(547600)_{10}$

$$(r^n - 1) - N$$

n:= no. of items
N:= no itself

$$((10)^6 - 1) - 547600 = 452399$$

Radix complement

$(1234)_{10}$ $\left[(r^n-1)-N\right]+1$

$\left[((10)^4-1)-1234\right]+1 = \boxed{8766}$

Radix complement

$(12004)_5$ find $(r-1)$ complement

$(r^n-1)-N$

$((5)^5-1)-12004$

$(3124)_{10} - 12004$

converting to base 5

5	3124
5	624 - 4
5	124 - 4
5	24 - 4

44444 - 12004

= 32440

☆ note $(r^n - 1)$ will always give answer in decimal

$(03DBA)_{16}$ find (r-1) complement

$((16)^5 - 1) - 03DBA$

$(1048575)_{10} - 03DBA$

16	1048575
16	65535 - 15
16	4095 - 15
16	255 - 15
	15 - 15

$(FFFFF)_{16} - (03DBA)_{16}$

$= 16\ 15\ 15\ 15\ 15$

$-\ 0\ 3\ 13\ 11\ 10$

$0\ 12\ a\ 4\ 5$

$\rightarrow (FC245)_{16}$

finding r's complement

$(042142)_5$

$$\left[(r^n - 1) - N\right] + 1$$

$$\left[((5)^6 - 1) - 042142\right] + 1$$

$$\left[15624 - 042142\right] + 1$$

5	15624
5	3124 – 4
5	624 – 4
5	124 – 4
5	24 – 4
	4 – 4

444444

$$\left[444444 - 042142\right] + 1$$

$$= 402303$$

taking complement of a complement restores the original number.

fractions (r-1)

$(1101111 \cdot 1101)_2$ Remove point

$(11011111101)_2$

$((2)^{10} - 1) - 11011111101$

$\Rightarrow (1023) - 1101111101$

$\Rightarrow$

512	256	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	1	1

$(1111111111)_2$

$$\begin{array}{r} 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1 \\ -\ 1\ 1\ 0\ 1\ 1\ 1\ 1\ 1\ 0\ 1 \\ \hline (0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 0)_2 \end{array}$$

$\rightarrow (0010000.0010)_2$

r's complement

$(0723.6124)_8 = (07236124)_8$

$$\left[(r^n - 1) - N\right] + 1$$

$$\left((8)^8 - 1\right) - 07236124$$

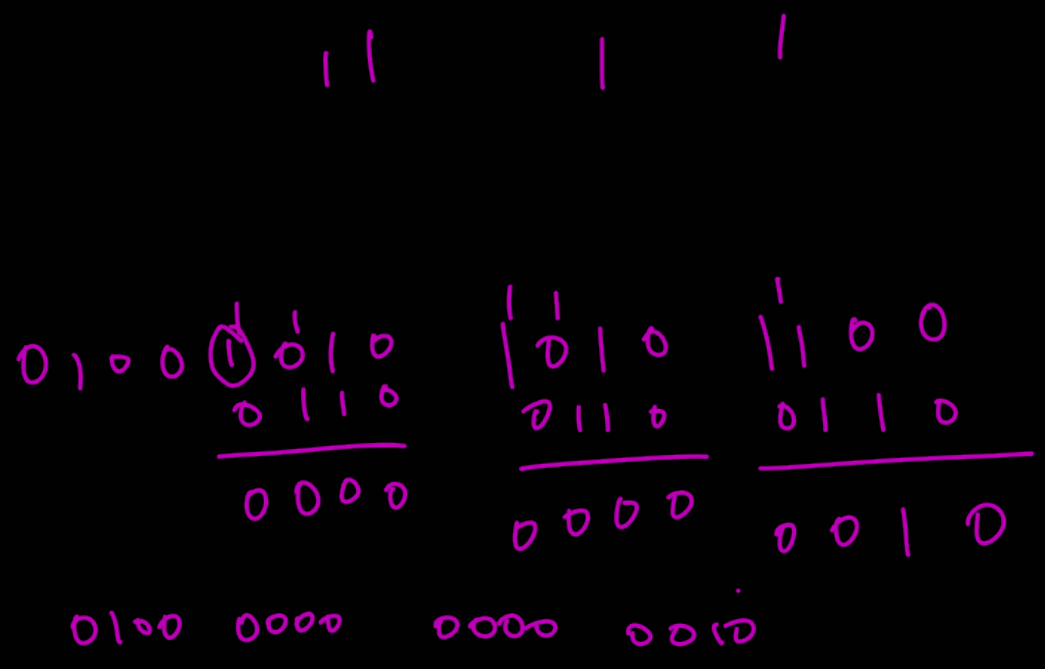
$$(16777215)_{10} - (07236124)$$

8	16777215
8	2097151 −7
8	262143 −7
8	32767 −7
8	4095 −7
8	511 −7
8	63 −7
	7 −7

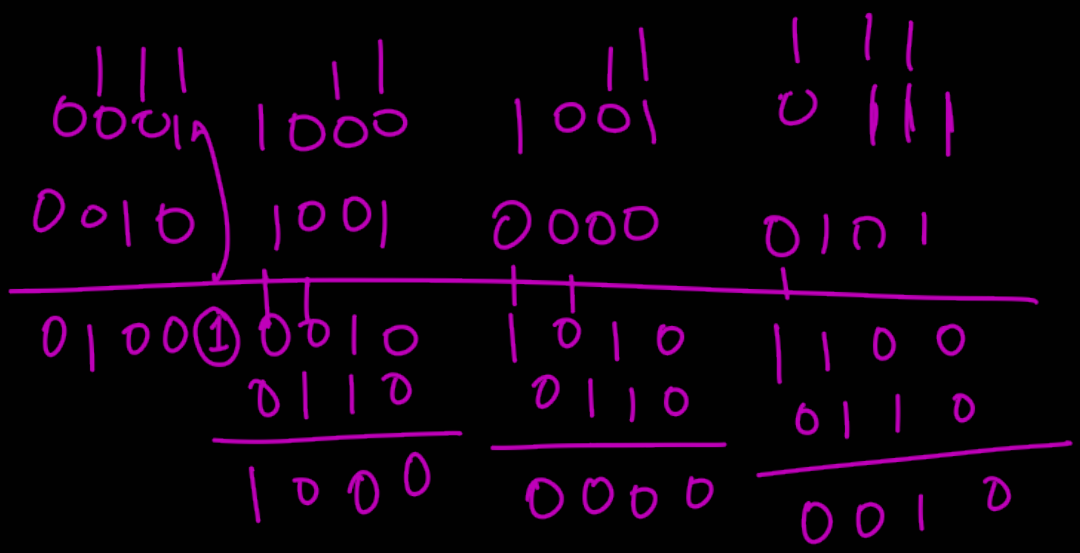
$(77777777)_8 - (07136124)_8$

$= 70541653 + 1$

$= \underline{(7054.1654)_8}$



0100 0000 0000 0010



0100 1000 0000 0010

Octal Addition

Range (0-7)

$$\begin{array}{r} {\scriptstyle 1\ \ 1} \\ 23 \\ +\ 67 \\ \hline (112)_8 \end{array}$$

8	10
	1-2

→ decimal

$$8\overline{)10}\ \begin{array}{l}1\\ -8\\ \hline 2\end{array}$$

$(10)_{10} \rightarrow (12)_8$

8	9
	1-1

$$8\overline{)9}\ \begin{array}{l}1\\ -8\\ \hline 1\end{array}$$

$$\begin{array}{r} {\scriptstyle 1\ \ 1} \\ 7\ 6\ 02 \\ +\ 5\ 7\ 7\ 1 \\ \hline 15\ 5\ 7\ 3 \end{array}$$

8	13
	1-5

$$8\overline{)13}\ \begin{array}{l}1\\ -8\\ \hline 5\end{array}$$

$(15573)_8$

8	13
	1-5

Octal Subtraction

Borrow = 8 from next column

$$\begin{array}{r} {}^{1}\not{2}\ {}^{8+4}\not{5}\ {}^{8+}3 \\ -\ 1\ 6\ 7 \\ \hline 0\ 6\ 4 \end{array}$$

$253 - 167 = (64)_8$

$$\begin{array}{r} 2\ {}^{2}\not{3}\ {}^{8+1}0\ {}^{8+}4 \\ -\ 1\ 0\ 6\ 5 \\ \hline 1\ 2\ 1\ 7 \end{array}$$

$(1217)_8$

Hexadecimal Addition

$$\begin{array}{r} {\scriptstyle 1}\quad {\scriptstyle 1} \\ 7\ A\ 6 \\ +\ 2\ B\ A \\ \hline 10\ 6\ 0 \end{array}$$

$(A60)_{16}$

$$\begin{array}{r} {\scriptstyle 1}\quad {\scriptstyle 1} \\ 7\ F\ 5 \\ 5\ B\ D \\ \hline 13\ B\ 2 \end{array}$$

$\rightarrow (DB2)$

16	18
	1-2

16	27
	1-11
	1-B

Hexadecimal Subtraction

Borrow = 16

$$\begin{array}{r} 6 \;\; \overset{16+B}{\not 7} \;\; B \;\; 6 \\ - \;\; 6 \;\; C \;\; 5 \\ \hline 0 \;\; 15 \;\; 1 \end{array}$$

$$= (F1)$$

$$\begin{array}{r} F \;\; \overset{\&}{\not 9} \;\; \overset{16+}{A} \;\; C \\ - \;\; D \;\; 4 \;\; C \;\; 7 \\ \hline 2 \;\; 4 \;\; 14 \;\; 5 \end{array}$$

(24E5)

gray code → one bit changes so min error.

(un-weighted)

Binary to gray code conversion MSB remains same.

Deci	Binary	Gray Code
2	10 ↔ XOR	11

XOR
Same - 0
Diff - 1

logic gates



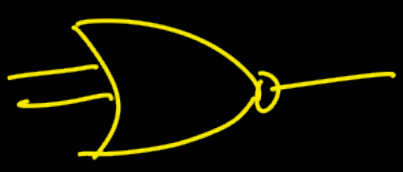
And (multiplication) $A \cdot B$



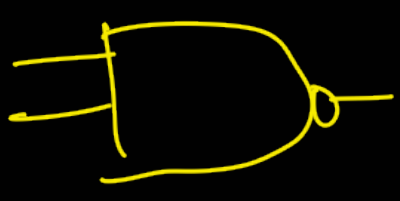
OR (add) $A+B$



Not (inverse) $\overline{A}$



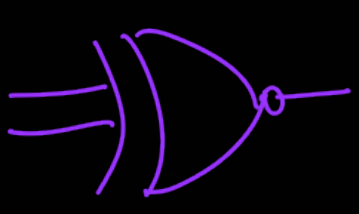
NOR (inverse of OR) $\overline{A+B}$



NAND (inverse of AND) $\overline{A \cdot B}$



XOR Same → 0, diff → 1 $A \oplus B$



XNOR flip output of XOR $A \odot B$

literal = generic term used for Variable or its complement. A, A', y'

no of Inputs in a circuit determine no of Rows of truth table. 2^{inputs} :- ?

Important Rules

$A+0=A$ $\quad A\cdot A=A$

$A+1=1$ $\quad A\cdot\overline{A}=0$

$A\cdot 0=0$ $\quad \overline{\overline{A}}=A$

$A\cdot 1=A$ $\quad A+AB=A$ $\quad A\cdot(1+B)=A\cdot(1)$

$A+A=A$

$A+\overline{A}=1$ $\quad A(B+C)=AB+AC$

De morgan law

$$\overline{A\cdot B}=\overline{A}+\overline{B}$$

$$\overline{A'B\cdot C}:\overline{A}+\overline{B}+\overline{C}$$

Sum term:: The Sum of two or more literals.

$A+B'$, $x+y$

Product term

The product of two or more literals

AB' , $w'xyz$

BCD → 4 bit representation

no base

10 BCD code will be Seperate (4 bits for each digits)

$(0001\,0000)_2$

if Range is greater than = 10 add 6

(0-9)

$(0110)_2$

BCD Addition

4 → 8 4 2 1
0 1 0 0

8 → 8 4 2 1
1 0 0 0

i) 4 + 8

```
  1000
+ 0100
------
  1100   →
+ 0110
------
① 0010   →  0001 0010  →  12
```

check 8 4 2 1
1 1 0 0 → 8+4 = 12
exceeds 9
So add 6

final Answer

ii) Solve

184 + 576

184 → 0001 1000 0100

576 → 0101 0111 → 0110

```
  1  11  1  1
 0001 1000 0100
 0101 0111 0110 +
-----------------
 0111 0000 1010
    0110
+ ------
 0111 0110
+ 0110
-------
 0111 0110 0000
```

→ 0111 0110 0000

After every addition first check its Range. and add 0110 if needed. Then move on.

Carry to the next term is only done after 0110 has been added.

BCD Subtraction

e.g 375 − 240 = 135

375 → 0011 0111 0101

240 → 0010 0100 0000

In Subtraction always Subtract 0110

```
  0011 0111 0101
− 0010 0100 0000
-----------------
 0001 0011 0101
```

$(1\ 3\ 5)_2$

e.g 206 − 147 → 0111

206 → 0010 0000 0110

147 → 0001 0100

(Parity Bit) if no. of ones is even = even parity.
if no. of ones is odd = odd parity.

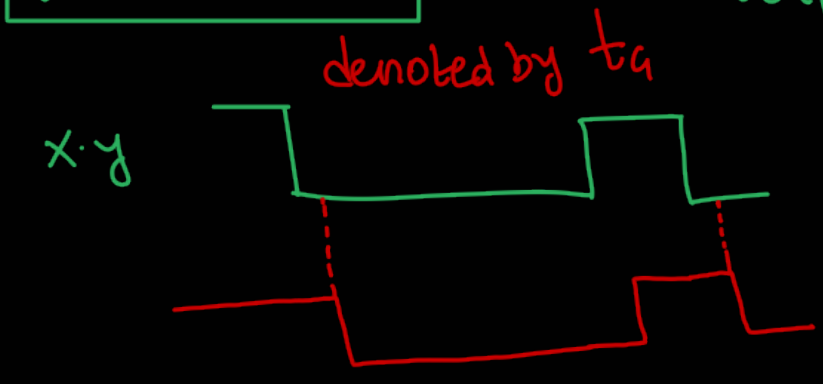
↓

use for communication errors.

ASCII → Seven Bit code:

But an extra bit is added to ASCII for again Parity check

gate delay time it takes to generate an output!



Reasons of t_g
→ no. of inputs
→ circuit design

Important Boolean Identities.

$$x + yz = (x+y)(x+z)$$

$$\overline{x+y} = \bar{x}\cdot\bar{y} \qquad \overline{x\cdot y} = \bar{x} + \bar{y}$$

↳ De morgan law

Duality of a function :

Replace all AND's with OR and vice versa.

Replace all 1's with 0's and vice versa.

$$F_d = A'B + B'$$

$$= \overline{(A' + B) \cdot B'}$$

F_d -> duality of a function.

e.g

$$xy + x\bar{y} \quad 2$$

$$(x+y) \cdot (x + \bar{y})$$

Demorgan law can be applied to any no. of variables.

$$\overline{x \cdot y \cdot z} = \bar{x} + \bar{y} + \bar{z}$$

$$\overline{x + y + z} = \bar{x} \cdot \bar{y} \cdot \bar{z}$$



Simplify.

$$\overline{(A + B \cdot C) \cdot (A \cdot C + B)} =$$

$\overline{x \cdot y} = \bar{x}1$

$$\overline{(A + B \cdot C)} + \overline{(A \cdot C + B)} =$$

$$(\bar{A} \cdot \overline{B \cdot C}) + (\overline{A \cdot C} \cdot \bar{B}) =$$

$$\bar{A} \cdot B\bar{C} + \overline{A \cdot C} \cdot \bar{B} =$$

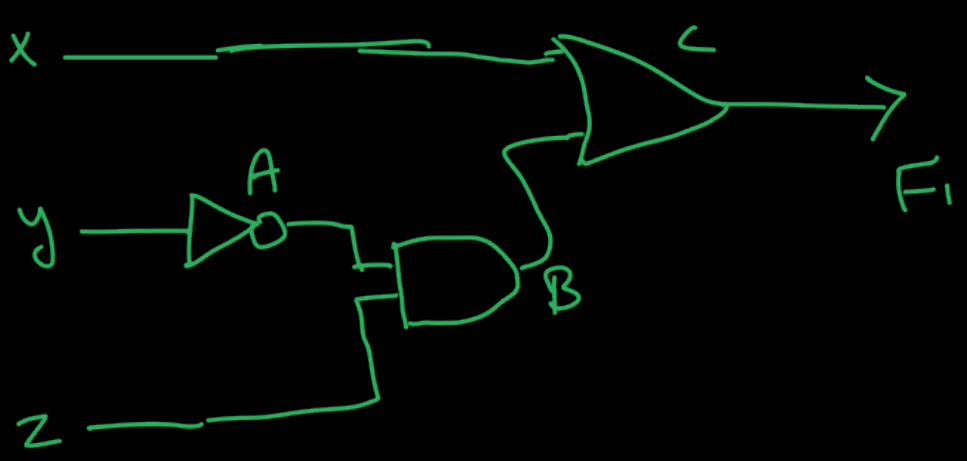
$$\bar{A} \cdot (\bar{B} + \bar{C}) + (\bar{A} + \bar{C}) \cdot \bar{B} =$$

$$\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B} + \bar{B}\bar{C} =$$

$$\boxed{\bar{A}\cdot\bar{B} + \bar{A}\cdot\bar{C} + \bar{B}\cdot\bar{C}}$$

no. of rows in a truth table $= 2^n$ where n is the no. of variables.

$$F_1 = x + y'z$$



$2^3 = 8$

x	y	z	A	B	C
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	0	1
1	1	0	0	0	0
1	1	1	0	0	1

Simplify $F_2 = x'y'z + x'yz + xy'$

$= \bar{x}\bar{y}z + \bar{x}yz + x\bar{y}$

$= \bar{x}z(\bar{y} + y) + x\bar{y}$

$= \bar{x}z(1) + x\bar{y}$

$= \boxed{\bar{x}z + x\bar{y}}$ → Simplified.

Canonical form

Sum of Products

$xyz + xy \longrightarrow$ SOP

↑ Products ↑ Products

Product of Sums

$(x+y+z)(y+z) \longrightarrow$ POS

↓ Sums ↓ Sum

Minterm A product term in which all variables appear exactly once either complemented or uncomplemented.

xy , $\bar{y}x$, $\bar{x}y$, $\bar{x}\bar{y}$ $\longleftarrow$ $xy \longleftarrow 2^2$

It has a value of 1 in truth table.

There are 2^n distinct min terms. $n \rightarrow$ no. of variables

minterms are donated by m

Note:
A literal is a complimented variable if its value is 0 and uncomplimented if its value is 1.

x	y	Min terms	Symbol
0	0	$\bar{x} \cdot \bar{y}$	m_0
0	1	$\bar{x} \cdot y$	m_1
1	0	$x \cdot \bar{y}$	m_2
1	1	$x \cdot y$	m_3

Complemented $\Longrightarrow 0$

Un-complemented $\Longrightarrow 1$

Solving minterms

$\bar{x} \cdot \bar{y} = \bar{0} \cdot \bar{0} = 1 \cdot 1 = (1)$

Maxterms: A Sumterm in which all the variables appear exactly once. either complemented or uncomplement.

Symbol: M

It has a 0 valve in $\overline{T}\,\overline{T}$

(x and y) Maxterms

2^N : N = variables

2^N : 2^2 : 4

- $x + y$
- $\bar{x} + y$
- $x + \bar{y}$
- $\bar{x} + \bar{y}$

In maxterm a complemented variable = 1 and

Uncomplement variable = 0

x	y	Maxterms	Symbol
0	0	$x + y$	M_0
0	1	$x + \bar{y}$	M_1
1	0	$\bar{x} + y$	M_2
1	1	$\bar{x} + \bar{y}$	M_3

Solving.

$x + \bar{y} = 0 + \overline{T} = \boxed{0}$

note Relationship b/w minterm and maxterm

They are complements of each other with same subscripts e.g $\bar{J}=3$

$$M_J = \overline{m_J} \qquad m_J = \overline{M_J}$$

(find the max terms)

x	y	z	F	
0	0	0	1	M_0
0	0	1	0	M_1 $x+y+\bar{z}$
0	1	0	1	M_2
0	1	1	0	M_3 $x+\bar{y}+\bar{z}$
1	0	0	0	M_4 $\bar{x}+y+z$
1	0	1	1	M_5
1	1	0	0	M_6 $\bar{x}+\bar{y}+z$
1	1	1	1	M_7

Maxterm ↓ always written in POS. t.t value $\Rightarrow 0$

final Answer.

$$F = (x+y+\bar{z}) \cdot (x+\bar{y}+\bar{z}) \cdot (\bar{x}+y+z) \cdot (\bar{x}+\bar{y}+z)$$

↓ expression form

$$= M_1 \cdot M_3 \cdot M_4 \cdot M_6$$

$$F(x,y,z) = \prod M(1,3,4,6)$$

↓

Logical product

Complement of a function

When function F expressed As Sum of Min terms are requirement is to find the expression in max-terms form. Step: Select min terms missing in Sum of min terms form

Max no is 7 hence
$m_0 \rightarrow m_7$

$$F = \sum m(1,3,5,7)$$

Note down the missing min terms.

$$\overline{F} = \sum m(0,2,4,6)$$

Example: For a given function $F(x,y,z)$ below find max terms.

$$F(x,y,z) = \sum_m (1,3,5,7)$$

this means these are the values where output is 1. Now finding the missing minterms of this means we are getting values where output is 0 aka Maxterms.

$$\overline{F(x,y,z)} = \sum m(0,2,4,6)$$

$$= m_0 + m_2 + m_4 + m_6$$

Hence we firstly took a complement.

Now again taking complement

$$F(x,y,z) = \overline{m_0 + m_2 + m_4 + m_6}$$

$$= \overline{m_0} \cdot \overline{m_2} \cdot \overline{m_4} \cdot \overline{m_6}$$

Since $\overline{m_J} = M_J$

$$= M_0 \cdot M_2 \cdot M_4 \cdot M_6$$

$$= F(x,y,z) = \prod M(0,2,4,6)$$

To convert in POS expression, write the Binary equivalent of J

M(0, 2, 4, 6) → 000, 010, 100, 110

1 → complemented
0 → uncomplemented

1 1 0 → x $\bar{y}$ $\bar{z}$

$$(x+y+z)(x+\bar{y}+z)(\bar{x}+y+z)(\bar{x}+\bar{y}+z)$$

Two Standard forms

→ Sum of Product (SoP) — Sum of minterm

→ Product of Sum (PoS) — Product of max term

> Same thing

Simplication of Sum of minterms leads to Sop.

Same for Product of maxterms Leads to Pos.

Sop $\xleftarrow{\text{Complement}}$ Pos

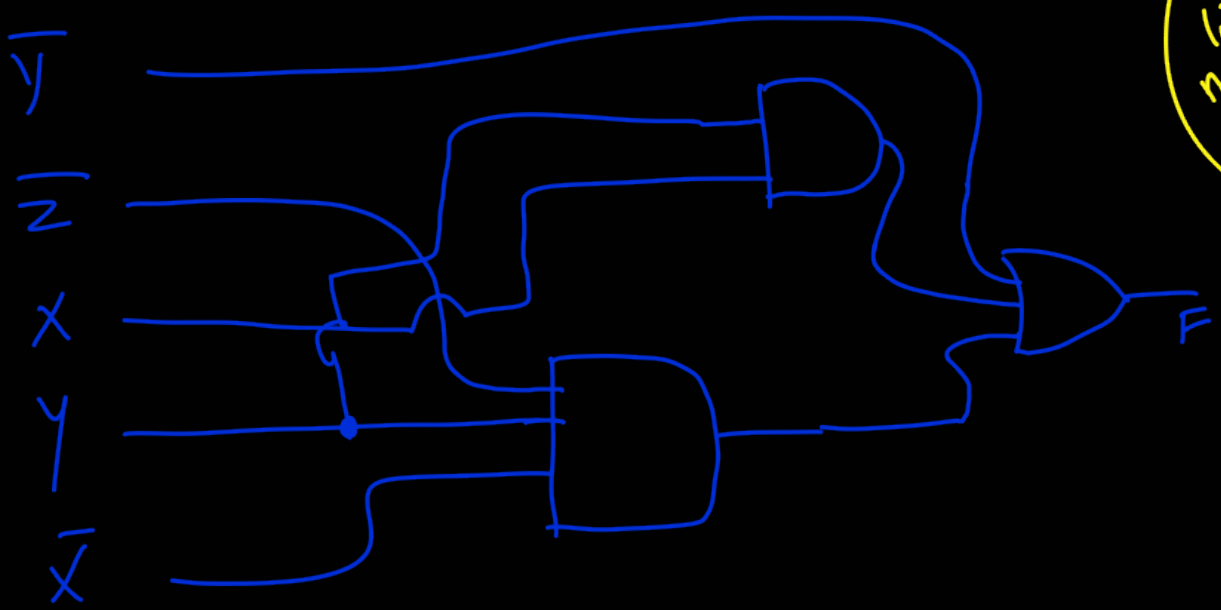
SoP|PoS is Simplified where literals can appear or not.

Sop

e.g $F = \bar{y} + \bar{x}y\bar{z} + xy$

Requires 2 And gate and 1 OR

But in minterms/maxterms literals must appear in all



no need of not gate as variables are directly considered complemented / uncomplemented.

This function is not in SOP form.

$$F = AB + C(D + \bar{E})$$

$$F = AB + CD + C\bar{E}$$

POS

$$F = \bar{A} \cdot (A + \bar{B} + C) \cdot (\bar{B} + \bar{C} + D)$$

$$F = (\text{Sum term})_1 \cdot (\text{Sum term})_2 \cdot (\text{Sum term})_3$$

$$F = \bar{y} + \bar{x} \cdot \bar{z}$$

is a Sum of Product But not minterm

A minterm is a Single Product exp.
A maxterm is a Single Sum exp.

Minterm $A\bar{B}C\bar{D}$ ⇐ Single expression

Som $(\bar{A}B\bar{C}\bar{D}) + (\bar{A}BC\bar{D}) + (ABCD)$

Sop $AB + ACD$ ⇐ Simplified

SSOP $(\bar{A}\bar{B}CD) + (A\bar{B}\bar{C}D)$ ⇐ All variables present

Domain of Boolean expression.

↓

Set of Variables contain in expressions, either complemented or uncomplemented.

$$ABC + C\bar{D}E + \bar{B}C\bar{D}$$

Domain = A, B, C, D, E = 5 literals.

Standard Sum of Product (SSOP)

In SOP some of the term don't contain all variables e.g $y + (x \cdot \bar{y})$

But in SSOP every single term contains every variable present in that e.g.

e.g $(A \cdot B \cdot \bar{C} \cdot D) + (\bar{A} \cdot \bar{B} \cdot \bar{C} \cdot D)$

if domain = 4 literals.

$$\bar{A}BCD + ABCD + \bar{A}BCD$$

(SPOS) → again every single term contain all variables.

POS → not necessary that every term contains every variable.

SPOS e.g $(A + \bar{B} + C + \bar{D}) \cdot (\bar{A} + \bar{B} + \bar{C} + \bar{D})$

SSOP ⟷ SPOS

Complement

SSOP | SPOS heyps in K-maps.

converting SOP → SSOP

→ multiply each term that is missing a variable with $(x + \bar{x})$ Note $(x+\bar{x}) \rightarrow 1$

e.g $A \cdot \bar{B} \cdot C + \bar{A} \cdot \bar{B} + A \cdot B \cdot \bar{C} \cdot D$

Domain → A B C D

$(A \cdot \bar{B} \cdot C)(D+\bar{D})$

$A\bar{B}CD + A\bar{B}C\bar{D}$

$(\bar{A} \cdot \bar{B})(C+\bar{C})$

$[\bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}]$ fill this eq with missing variable

$(\bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C})(D+\bar{D})$

$\bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$

combine all

$A\bar{B}CD + A\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$

$+ AB\bar{C}D$

SSOP

Converting POS → SPOS

Add $(x \cdot \bar{x})$ to the term that is missing the variable.

$$x \cdot \bar{x} \rightarrow 0$$

e.g $(A + \bar{B} + C) \cdot (\bar{B} + C + \bar{D}) \cdot (A + \bar{B} + \bar{C} + D)$

$(A + \bar{B} + C) + (D \cdot \bar{D})$

↓ ↓

A BC

Apply distributive law

$$A + BC = (A + B) \cdot (A + C)$$

$$(A + \bar{B} + C + D) \cdot (A + \bar{B} + C + \bar{D})$$

Second term

$$\bar{B} + C + \bar{D} + (A \cdot \bar{A})$$

☆ $(A + BC) = (A + B) \cdot (A + C)$

$$(A + \bar{B} + C + \bar{D}) \cdot (\bar{A} + \bar{B} + C + \bar{D})$$

Converting SSOP → SPOS

e.g $\bar{A}\cdot\bar{B}\cdot\bar{C} + \bar{A}\cdot B\cdot\bar{C} + \bar{A}\cdot B\cdot C + A\cdot\bar{B}\cdot C + A\cdot B\cdot C$

write the Binary equivalent for each term.

$\bar{A}\cdot\bar{B}\cdot\bar{C} + \bar{A}\cdot B\cdot\bar{C} + \bar{A}\cdot B\cdot C + A\cdot\bar{B}\cdot C + A\cdot B\cdot C$

$000 + 010 + 011 + 101 + 111$

$0 \quad 2 \quad 3 \quad 5 \quad 7$

This SSOP contains 5 combinations. We have $2^3 = 8$ total combinations. Remaing combinations are 3 which are for SPOS.

Remaining

001 100 110 → now write in SPOS

$(x+y+\bar{z})\cdot(\bar{x}+y+z)\cdot(\bar{x}+\bar{y}+z)$

MSB ← X Y Z ← Least Significant Bit:

X	Y	Z
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

if we change Position of x y z it changes output

$$\bar{X}yz + x\bar{y}\bar{z} + xz$$
$$1\ \ 2\ 3\ \ 4\ 5\ 6\ \ 7\ 8$$

how many literals:

→ 8

how many terms

→ 3

Boolean Identities:

$X+0=X$ $\quad X\cdot 1=X$

$X+1=1$ $\quad X\cdot 0=0$

$X+X=X$ $\quad X\cdot X=X$

$X+\bar{X}=1$ $\quad X\cdot\bar{X}=0$

$\bar{\bar{X}}=X$

$\overline{A+X+Y}=\bar{X}\cdot\bar{Y}\cdot\bar{A}$

$\overline{X\cdot Y}=\bar{X}+\bar{Y}$

☆

$X+YZ=(X+Y)(X+Z)$

e.g $\quad X+\bar{X}Y$

e.g $\quad X+1=1$

$(A\cdot B)+1=1$

$(X+\bar{X})(X+Y)$

$1(X+Y)=X+Y$

Note in dual operation we replace

AND → OR

0 → 1

1 → 0

OR → AND

$X+1=1$

$X\cdot 0=0$

} dual

e.g

$H = A \cdot B + A \cdot C + B \cdot C$

dual $H = A+B \cdot A+C \cdot B+C$

consensus theorem ☆

To apply
- → 3 no. of literals
- → each literal appear twice
- →

$$X \cdot Y + \bar{X} \cdot Z + Y \cdot Z = XY + \bar{X}Z$$

$$(X+Y) \cdot (\bar{X}+Z) \cdot (Y+Z) = (X+Y) \cdot (\bar{X}+Z)$$

Complement of a function.

dual ↙ ↘ de-morgan's Law

By using de-morgan law

$F_1 = \bar{x}y\bar{z} + \bar{x}\bar{y}z$

$\overline{F_1} = \overline{\bar{x}y\bar{z} + \bar{x}\bar{y}z}$

$\overline{F_1} = (\overline{\bar{x}y\bar{z}}) \cdot (\overline{\bar{x}\bar{y}z})$

$\overline{F_1} = (\bar{\bar{x}} + \bar{y} + \bar{\bar{z}}) \cdot (\bar{\bar{x}} + y + \bar{z})$

$\overline{F_1} = (x + \bar{y} + z)(x + y + \bar{z})$

e.g 2 $F_2 = x(\bar{y}\bar{z} + yz)$

$\overline{F_2} = \overline{x(\bar{y}\bar{z} + yz)}$

$\overline{F_2} = \bar{x} + \overline{(\bar{y}\bar{z} + yz)}$

$\overline{F_2} = \bar{x} + (\overline{\bar{y}\bar{z}} \cdot \overline{yz})$

$\overline{F_2} = \bar{x} + (yz \cdot (\bar{y} + \bar{x}))$

$\overline{F_2} = \bar{x} + (y \cdot \bar{y}z + \bar{x}yz)$

$\overline{F_2} = \bar{x} + \bar{x}yz$

$\overline{F_2} = \bar{x}(1 + yz)$

$\overline{F_2} = \bar{x}$

By using dual: Take complement of each literal and change

$* \rightarrow +$ and $+ \rightarrow *$

$$F_2 = x(\bar{y}\bar{z} + yz)$$

$$\bar{F}_2 = \bar{x} + (\bar{y}\bar{z} + yz)$$

$$\bar{F}_2 = \bar{x} + (y+z)\cdot(\bar{y}+\bar{z})$$

$$\boxed{\bar{F}_2 = \bar{x} + (y+z)(\bar{y}+\bar{z})}$$

Solve $J=3$ and using De-morgan law to take complement

$$m_3 = \overline{\bar{x}\cdot y\cdot z}$$

$$= \bar{\bar{x}} + \bar{y} + \bar{z}$$

$$= x + \bar{y} + \bar{z} = m_3$$

x	y	z	
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
→ 0	1	1	m_3
1	0	0	
1	0	1	
1	1	0	
1	1	1	



Buffer gate:

x	F
0	0

$(A+B)(\bar{A}+C)$

$A(\bar{A}+C)+BL\bar{A}+C)$

$0+AC+\bar{A}B+BL$ — 3 literals — Appear twice

$AC+\bar{A}B+BL$ — Remain term

By using Consensus theorem.

Remaining term → That contains a literal which is both in its complemented and uncomplement form.

Discard the other term here $\bar{A}B$ and Ac remain.

$$AC+\bar{A}B$$

Reduce to 4 literals.

$(A+B)(A+CD)$

$A(A+CD)+B(A+CD)$

$A+ACD+AB+BCD$

$A+AB+ACD+BCD$

$A(1+AB)+ACD+BCD$

$A+ACD+BCD$

$A(1+CD)+BCD$

$A+BCD \rightarrow$ 4 literals reduced

Implementing function using AND, OR gates.

Q Express in OR gate only and complement operation.

$$F = A \cdot \overline{B} \cdot C + \overline{A} \cdot \overline{C} + A \cdot B$$

Taking double complement as it does not change the result.

$$\overline{\overline{F = A \cdot \overline{B} \cdot C + \overline{A} \cdot \overline{C} + A \cdot B}}$$

$$\overline{F} = \overline{(A \cdot \overline{B} \cdot C) \cdot (\overline{A} \cdot \overline{C}) (\overline{A} \cdot B)}$$

$$F = \overline{(A + \overline{\overline{B}} + \overline{C}) \cdot (\overline{\overline{A}} + \overline{\overline{C}}) \cdot (\overline{A} + \overline{B})}$$

$$F = \overline{(A + B + \overline{C}) \cdot (A + C) \cdot (\overline{A} + \overline{B})}$$

$$F = \overline{(A + B + \overline{C})} + \overline{(A + C)} + \overline{(\overline{A} + \overline{B})}$$

Don't open complement now as we will get dot.

$$F = \overline{(A + B + \overline{C})} + \overline{(A + C)} + \overline{(\overline{A} + \overline{B})}$$

final Answer.

Use this $(F = \overline{\overline{F}})$

Cost Criteria

Literal cost: The no. of literal appearances in a boolean function. A literal that is complemented OR uncomplement is both counted in literal cost

e.g

$F = \underset{1\ 2}{BD} + \underset{3\ 4\ 5}{A\bar{B}C} + \underset{6\ 7\ 8}{A\bar{C}\bar{D}}$ literal cost = 8

Calculating gate cost → total no of gates used in logic diagram

OR

For SOP and POS equations gate cost can be found out by

1) counting all literal appearances
2) no. of terms excluding terms with single literal
3) no. of Distinct complemented single literals.

e.g

$F = BD + A\bar{B}C + A\bar{C}\bar{D}$

literal = 8

no. of terms = 3

Distinct complement = 3

gate cost = $8 + 3 + 3$

$= 14$

e.g

$F = BD + A\bar{B}C + \bar{A}\bar{D}\bar{D} + AB\bar{C}$

gate cost = $4 + 11 + 3$

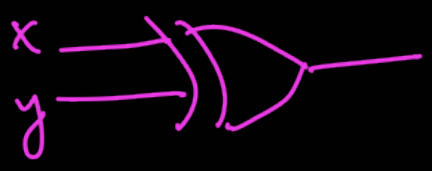
e.g

$F = (A + B')(A + D)(B' + C + \underline{D})(\underline{B}' + \underline{C} + D)$

gate cost = $10 + 3 + 4$

(XOR and xnor gates)

→ used to reduce the complexity of a circuit hence which reduces the cost of circuit



Same : 0
diff : 1

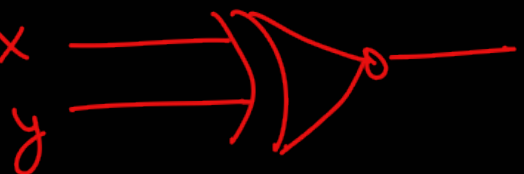
$$F = x\bar{y} + \bar{x}y = x \oplus y$$

note odd numbers of 1's on input side = 1 on output side

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

← only 1 hence odd

X NOR



Same : 1
diff : 0

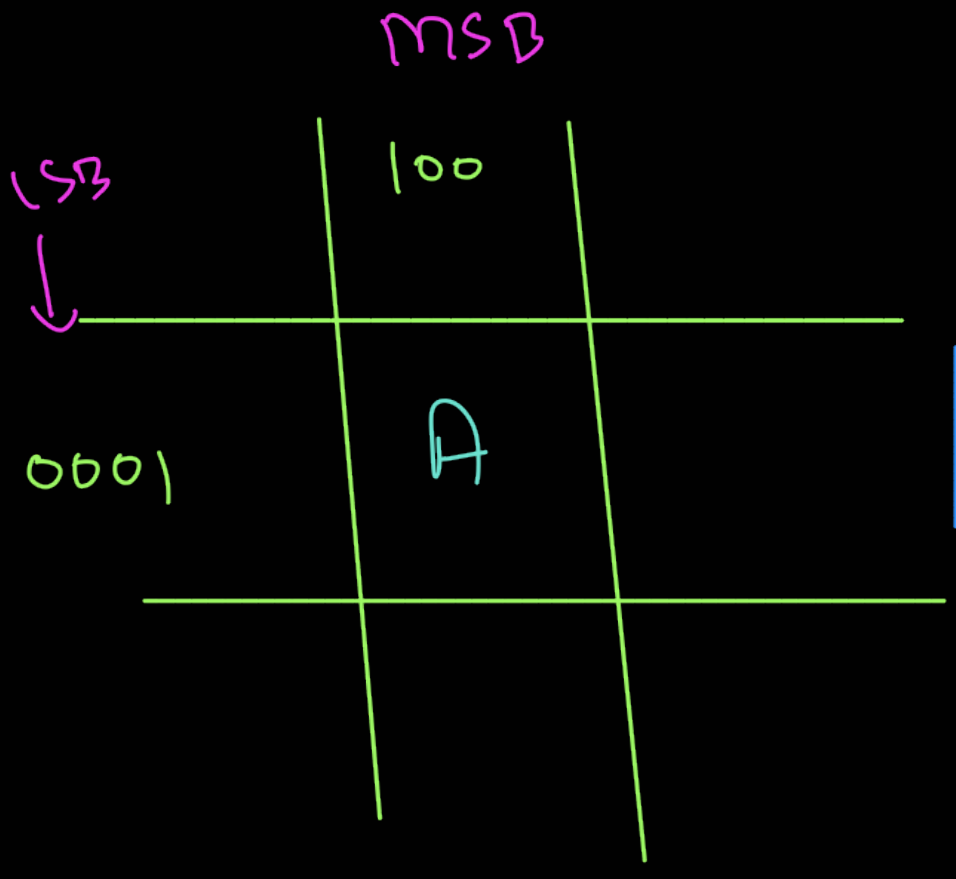
$$F = xy + \bar{x}\bar{y} = \overline{x \oplus y}$$

note: if we have even numbers of 1 in input then output → 1.

x	y	F
0	0	1
0	1	0
1	0	0
1	1	1

not 1's $\Rightarrow$ 0 hence even

How to Read ASCII



Find ASCII code of A.

100 0001

The MSB is determined by column. and the LSB is determined by row.

Representing Binary numbers.

How many distinct no.s can be represented in 'n' bits.

Possible state in Binary

0 1

Possible combinations $= 2^n$

n → no. of bits

E.g for 4 bits $\Rightarrow 2^4 = 16$ combinations



un-Signed Binary.

Range

0 to $(2^n - 1)$

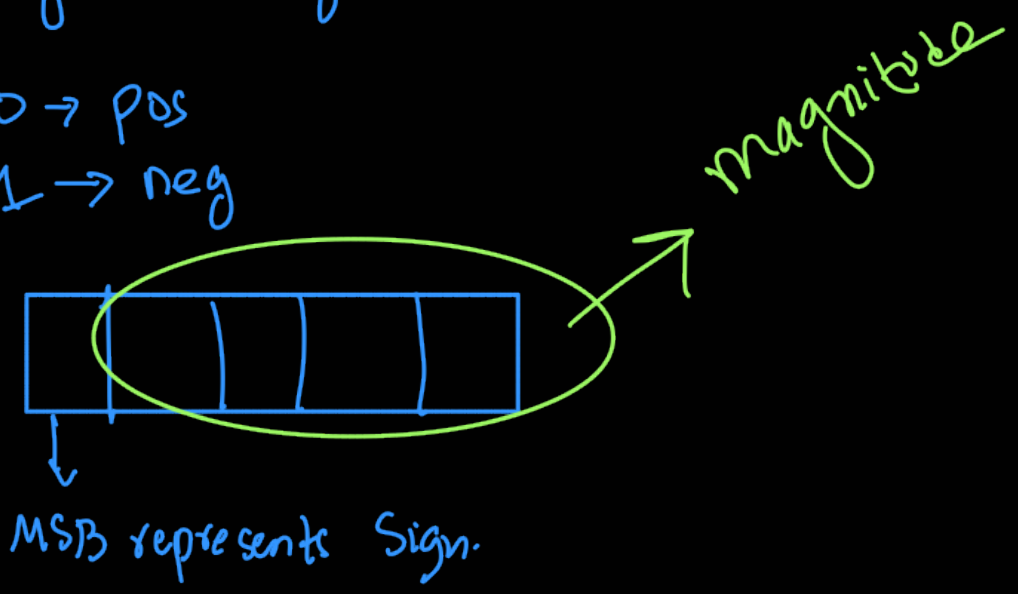
n = no. of bits

Range of Sign and mag method

$-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$

Sign and Mag Method

0 → pos
1 → neg



note in 1's complement
MSB represents Sign

Range of 1's complement
is Same as that of
Sign and magnitude

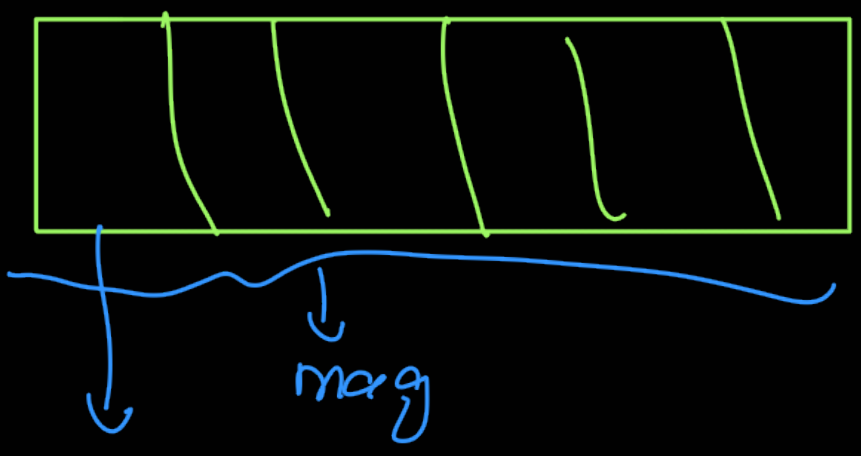
$-(2^{n-1}-1)$ to $+(2^{n-1}-1)$

(1's complement)

0000 ⟶ 1111

+0 -0

note in 2's complement it includes the sign in magnitude itself.



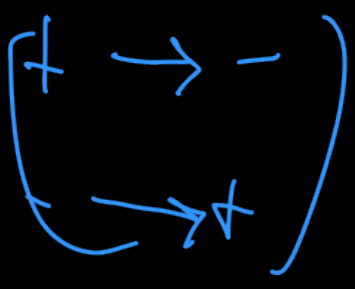
(MSB is Sign)

Range for 8's complement

$-(2^{n-1})$ to $+(2^{n-1}-1)$

By taking

2's complement



to verify Answer in

2's complement use

- weight if 1.

Bit extension in 2's

4bits (-7) → 1001

8bits (-7) → 1111 1001

if we have a negative number add 1's at the end.

How to perform Subtraction using 2's.

6 - 13 = ?

1) use un-signed to represent

8 4 2 1 4 2 1 → 1 1 0 → 6
1 1 0 1 1 1 0 → 1 1 0 1 → 13

2) Signed representation

6 → 4) 0 1 1 0 → +6
+13 → 4) 0 1 1 0 1 → +13

use positive only.

3) Make no. of bits same and check if decimal Ans is in range (6 - 13) ⇒ -7

00110 → +6

01101 → +13

4) Takes two complement on the no that is negative

+13 → -13

00110 → (+6)

+ 10011 → (-13)

Another e.g. $-6 - 13 = ?$

1) unsigned

$1\,1\,0 \rightarrow 6$

$1\,1\,0\,1$ ↓ 13

2) Signed

$0\,1\,1\,0 \rightarrow +6$

$0\,1\,1\,0\,1 \rightarrow +13$

) use positive identity

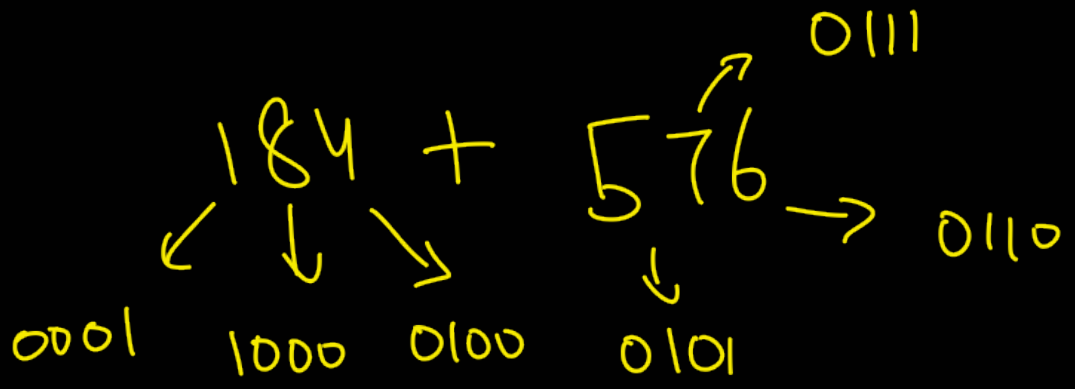
3) Check Range and extend bits

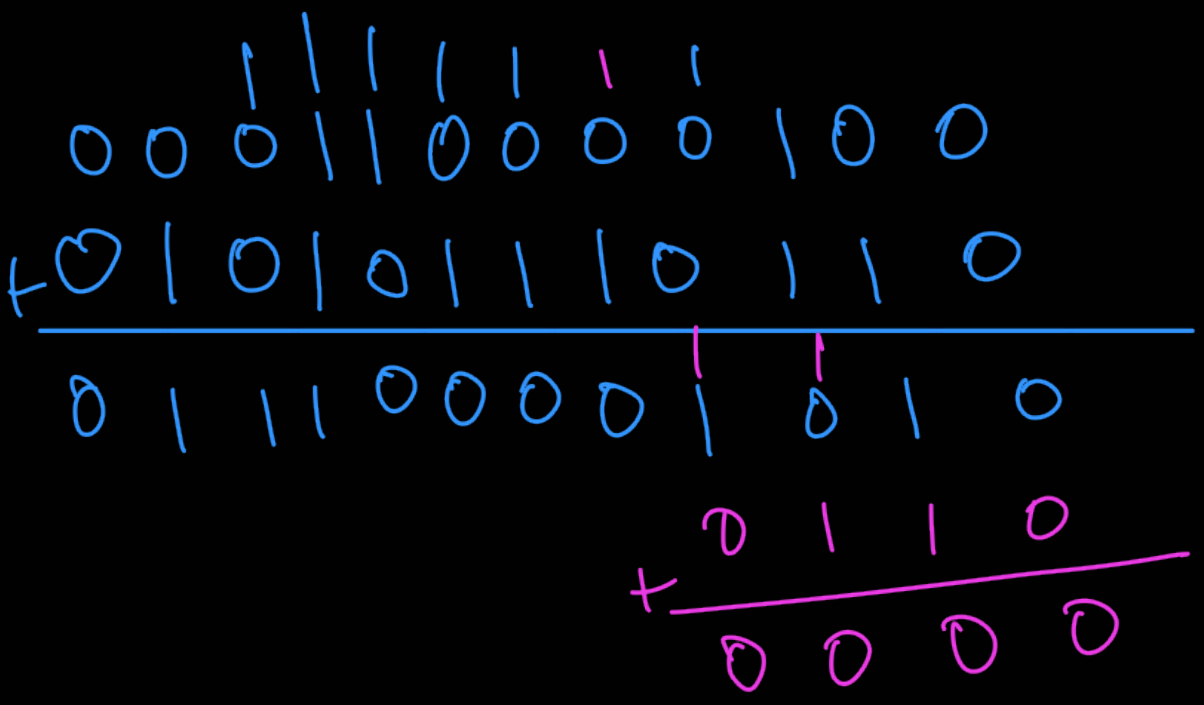
$+6 \rightarrow 0\,0\,0\,1\,1\,0$

$+13 \rightarrow 0\,0\,1\,1\,0\,1$

convert both to 2's complement then Add

Discard carry





0 111 0000 0 000